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Dip. in Engg. (Third Semester))/ P. T. D. C. (Fourth Semester) EXAMINATION, Dec.-Jan., 2013-14 (Elect.)

ELEMENTS OF ELECTRICAL ENGINEERING

Time : Three Hours] [*Maximum Marks : 100*

[*Minimum Pass Marks : 35*

Note : All questions are compulsory.

1. Solve any *two* of the following : 5 each

   (a) Explain Ohm's law and also factors affecting the resistance.

   (b) Explain the effect of temperature on resistance.

   (c) Find the resistance of 1000 mt. of a copper wire 25 mm^2 in cross-section. The resistance of copper is 1/58 Ω/mt length and 1 mm^2 cross-section. What will be the resistance of another wire of the same material three times as long and one half the cross-sectional area ?

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2. Solve any *three* of the following : 5 each

   (a) Classify insulator, semiconductor and conductor on the basis of energy band diagram.

   (b) What is insulating materials ? Write its properties.

   (c) Explain magnetic material and types of magnetic material.

   (d) What is semiconducting material ? Write its properties.

3. Numerical compulsory and solve any *two* questions from theory :

   (a) Derive expression for capacitance of parallel plate capacitor. 5

   (b) Explain the types of capacitor. 5

   (c) Derive expression for energy stored in a capacitor. 5

   (d) A 2 μF capacitor is connected by closing a switch to a supply of 100 V through a 1 M Ω series resistance. Calculate (i) time constant (ii) initial charging current (iii) the initial rate of rise of p. d. across capacitor (iv) voltage across capacitor 6 seconds after the switch has been closed and (v) the time taken for the capacitor to fully charged ? 10

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4. (a) What is inductor ? Explain the types of inductor. 5

Or

Derive expression for energy stored in a magnetic field.

(b) The field winding of a d.c. machine takes a steady current of 10 A, when connected to a d.c. source of 230 volts. On switching the voltage to the field, it is observed that it takes 0·3 seconds to reach 5 A : 10

(i) Find the value of inductance of the field winding.

(ii) Also find time for the current to fall to 30% of final steady value if the field terminal are short circuited.

5. (a) Write similarities and dissimilarities between electric circuit and magnetic circuit. 5

(b) Define permeability, reluctance, magnetic leakage, leakage flux, leakage coefficient. 5

(c) A circular iron ring has a mean circumference of 1·5 m and a cross-sectional area of 0·01 mt^2. A saw cut of 4 mm wide is made in the ring. Calculate magnetizing current required to produce a magnetic flux of 0·8 mWb in the air gap if the ring is wound with a coil of 175 turns. Assume relative permeability of iron as 400 and leakage factor 1·25. 10

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Or

An iron ring of cross-sectional area 6 cm^2 is wound with a wire of 100 turns and has a saw cut of 2 mm. Calculate the magnetizing current required to produce a magnetic flux of 0·1 mWb. If mean length of magnetic path is 30 cm and relative permeability of iron is 470. 10

6. (a) Explain Faraday's law of electromagnetic induction. 5

(b) Derive expression for force on a current carrying conductor placed in a magnetic field. 5

(c) Explain Fleming's left hand and right hand rule. 5

(d) A wooden toroid of mean diameter 400 mm and cross-sectional area 400 mm^2 is uniformly wound with a coil of 1000 turns which carries a current of 2 A. Determine (i) self-inductance of the coil and (ii) the e.m.f. induced in the coil when the current is uniformly reduced to zero in 10 ms. 5

Or

An iron ring of cross-sectional area 800 mm^2 and of mean radius of 170 mm has two windings connected in series, one of 500 turns and the other of 700 turns if the relative permeability of iron is 1200. Find (i) self-inductance of each coil and (ii) mutual inductance assuming that there is no leakage.

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